

# DriveSubsystem: FSM and Hardware Example

This page zooms into one subsystem. It shows object containment, requests, states, and the drive hardware wrapper.

TeamXRobot requests drive behavior; DriveSubsystem stores the request.

The FSM chooses the active State during update(); the State tells DriveHardware what motor powers to apply.

## Requests from TeamXRobot

`drive(forward,  
strafe, rotate)`

`enableManualDrive()`

`enableHeadingHold()`

`disableDrive()`

## driveSubsystem: DriveSubsystem

contains request fields, one FSM object, and one DriveHardware wrapper object

### request fields

stored by DriveSubsystem

`requestedMode`

`requestedForward`

`requestedStrafe`

`requestedRotate`

### fsm: FSM

starts in DisabledDriveState

`disabledState`

Disabled

`manualState`

Manual

`headingState`

HeadingHold

### Behavior policy

Transitions check requestedMode.

Only one transition fires per update.

States call DriveSubsystem helper methods.

`stopDrive()`

`applyManualDrive()`

`applyHeadingHoldPlaceholder()`

### driveHardware: DriveHardware

contains four required DcMotorEx objects

`frontLeft`

`frontRight`

`rearLeft`

`rearRight`

`setMotorPowers(fl, fr, rl, rr)`

`stop()`

Dashed boxes are explanations or concepts. Solid boxes are object instances contained by DriveSubsystem or DriveHardware.